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Interpersonal needs and psychological pain: The role of brooding and rejection sensitivity

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ABSTRACT

The Interpersonal Theory of Suicide proposes that thwarted belongingness (TB) and perceived burdensomeness (PB) predict psychological pain and desire for suicide. Brooding may be a mechanism in explaining how TB and PB predict pain. The conceptual similarity between rejection sensitivity (RS) and TB suggests that individuals with high RS may be likely to experience psychological pain in the context of TB. To test this model, 155 college students completed measures of psychological pain, TB, PB, brooding, and RS. As predicted, brooding mediated the relations between both TB and PB and psychological pain. RS interacted with TB in predicting psychological pain.

KEYWORDS

Risk factors; rumination; suicide

Intense psychological pain is an important predictor of suicidal thoughts and behavior across multiple theories (e.g., Klonsky & May, 2015; Shneidmen, 1993). An array of terms,

including mental pain, psychache, psychic pain, psychological pain, emptiness, internal perturbation, and psychological quality of life, have all been used to refer to the same construct of an intolerable emotional pain that increases risk for suicide (Tossani, 2013). Shneidman (1993), for example, defined psychache as an acute state of intense psychological pain associated with feelings of guilt, anguish, fear, panic, angst, loneliness and helplessness. He argued that psychache is the root cause of suicide, resulting from failure to meet any of a multitude of psychological needs. Research suggests that psychological pain predicts suicidal ideation and behavior beyond the effects of depression and hopelessness (e.g., Troister & Holden, 2010).

Similarly, the Interpersonal Theory of Suicide (ITS; Joiner, 2005; Van Orden et al., 2010) posits that desire for suicide results specifically when interpersonal needs are not met. Failure to meet these interpersonal needs can result in states of thwarted belongingness (TB), conceptualized as loneliness and the absence of reciprocal caring relationships, and/or perceived burdensomeness (PB), the belief that one's existence is a liability to others. TB and PB, as well as their interaction, are thought to predict desire for suicide (Joiner, Ribeiro, & Silva, 2012; Van Orden et al., 2010). A growing literature supports the relationship between thwarted interpersonal needs and suicidal desire (see Ma, Batterham, Cleave, & Han, 2016 for a review). It is important to note that ITS differs from other theories centered on psychological pain as the cause of suicide in that ITS adds a component of acquired capability for suicide that specifies a pathway from the more prevalent desire for suicide to the rarer acts of suicide attempts (Van Orden et al., 2010).

Despite their conceptual link as predictors of suicidal ideation, little research has examined TB or PB as specific contributors to the development of psychological pain. Similarly, there is a paucity of research examining other psychological processes that might impact the respective effects of TB and PB on psychological pain. One study examining suicide risk in college students

found that TB and PB indirectly predicted suicide risk via psychological pain (Campos & Holden, 2015), providing preliminary evidence of a relation between these constructs. In addition, Campos and Holden (2015) identified one environmental variable, childhood experiences of parental rejection, that impacted the development of TB, PB, and later suicide risk. Specifically, childhood rejection predicted TB and PB, both directly and indirectly, through depressive symptoms. Although it is unclear how childhood experiences of parental rejection relate to later TB and PB, one possibility is the personality trait of rejection sensitivity (RS). Both parental and peer rejection have been found to predict RS, which in turn predicts negative psychological outcomes, such as borderline characteristics (Rosenbach & Renneberg, 2014) and depressive symptoms (Downey & Feldman, 1996).

RS is a cognitive-affective processing style characterized by anxiety about expected and perceived interpersonal rejection (Downey & Feldman, 1996). This tendency to perceive rejection when faced with ambiguous social situations is associated with depression (Downey & Feldman, 1996) and loneliness in early adolescents and young adults (London, Downey, Bonica, & Paltin, 2007; Watson & Nesdale, 2012). RS is believed to develop when one's desire to belong is repeatedly thwarted, leading to expectations of further rejection. This fear leads to negative affect, including distress and anger, and maladaptive behavior such as hostility and withdrawal that can ultimately lead to further rejection (Levy, Ayduk, & Downey, 2001).

RS is an established correlate of depression with strong conceptual links to IPTS, as RS and TB are both associated with a perceived unmet desire for interpersonal connection. RS could be an underlying trait that leads to TB through hostility and withdrawal; conversely longstanding experiences of TB could produce an anxious expectation of further rejection. Given these possible dynamics, we predict that the interaction of RS and TB would produce increased levels of

psychological pain beyond the individual contributions from RS and TB. In contrast, RS is not conceptually linked to believing one is a burden on others (PB) and therefore we would not expect RS to interact with PB in predicting psychological pain.

In addition to RS, the ruminative process of brooding could act as a cognitive mechanism that fuels the psychological pain associated with suicidality. Response Styles Theory (RST) posits that rumination, a repetitive cognitive focus on the causes, meanings, and consequences of one's depressed mood, predicts depression and other negative psychological outcomes (Nolen-Hoeksema, 1991). In a review of 12 studies, 11 found an association between rumination and increased suicidal ideation and behavior (Morrison & O'Connor, 2008). Brooding, a form of rumination characterized by a tendency to dwell on the negative consequences of one's depression (Treyner, Gonzalez, & Nolen-Hoeksema, 2003) may be specifically relevant to psychological pain. Kerkhof and van Spijker (2011) theorized that brooding inherently increases psychological pain, as it involves frequent rehearsal of negative and hopeless thinking and is often experienced as uncontrollable. Consistent with this perspective, a recent meta-analysis found a moderate, significant relation between brooding and suicidal ideation (Hedges' $g = 0.63$, $p < .001$) in cross-sectional and prospective studies (Rogers & Joiner, 2017), suggesting its relevance to theories of suicidal ideation and behavior. Surprisingly, we are aware of no studies investigating the role of brooding on the interpersonal needs specified by IPTS. However, psychological pain resulting from unmet interpersonal needs could be facilitated by brooding. Thus, we predicted that brooding about one's interpersonal distress, in the form of either TB or PB, would lead to psychological pain.

Taken together, we propose a cognitive model of psychological pain that builds on IPTS by adding RS and RST. Specifically, we propose direct effects of TB and PB in predicting

psychological pain. In addition, we hypothesize that brooding will serve as an intervening variable in the relations between psychological pain and both TB and PB. Finally, we hypothesize that RS will predict psychological pain directly and will moderate the residual relation between TB and psychological pain, but not PB and psychological pain, controlling for brooding.

Method

Participants

Participants ($N = 199$) were recruited through the undergraduate psychology participant pool at a mid-sized university. Participants had to be 18 years or older to participate. No other inclusion criteria were utilized. Of the original sample, 42 cases were deleted due to missing values. Two additional cases were removed due to univariate outlier status on age. The final sample consisted of 155 participants. The mean age of the sample was 19.91 ($SD = 1.75$). Most participants were European-American (84.3%), women (57.4%), and heterosexual (96.8%).

Measures

The Psychache Scale (PAS; Holden, Mehta, Cunningham, & McLeod, 2001) is a 13-item self-report scale developed to assess Shneidman's (1993) construct of psychological pain as a relevant domain of suicide risk. Items are rated on a 1 - 5 Likert scale and total scores range from 13 - 65. Higher scores indicate more severe psychological pain. When tested in a sample of undergraduate students, the PAS had strong internal consistency ($\alpha = 0.92$) and accuracy in predicting suicidal ideation and attempts (Holden et al., 2001). Comparable internal consistency was found in this study ($\alpha = 0.97$), with scores ranging from 13 - 64 ($M = 17.80$; $SD = 8.94$).

The Interpersonal Needs Questionnaire, 10-item version (INQ-10; Bryan, Morrow, Anestis, & Joiner, 2010) is a 10-item self-report measure assessing TB and PB as they relate to Joiner's (2005) IPTS model (5 items measure TB and 5 items measure PB). Items are rated on a 7-point Likert scale (1 = *not at all true for me*; 7 = *very true for me*), with higher scores indicating more severe perceptions of TB and PB. When completed by community adults the scales were internally consistent (TB: $\alpha = 0.85$; PB: $\alpha = 0.93$) and predicted suicidal symptoms (Bryan et al., 2009). Internal consistencies were similar in this sample (TB: $\alpha = 0.85$; PB: $\alpha = 0.92$). Scores ranged from 5 - 35 for TB ($M = 12.85$; $SD = 7.21$), and 5 - 34 for PB ($M = 7.01$; $SD = 4.49$), respectively.

The Brooding Subscale of the Ruminative Response Scale (RRS-B; Treynor et al., 2003) is 5 items measuring the tendency to dwell passively and reflect judgmentally on one's situation when upset. Items are rated on a scale of 1 (*almost never*) to 4 (*almost always*). Scores range from 5 - 20, with higher scores indicating more frequent rumination. When tested in a sample of community adults, adequate internal consistency was obtained ($\alpha = 0.77$; Treynor et al., 2003). Good internal consistency was also observed in the present college students ($\alpha = 0.86$), with scores ranging from 5 - 20 ($M = 9.33$; $SD = 3.70$).

The Rejection Sensitivity Questionnaire - Short Form (RSQ-SF; Downey & Feldman, 2013) includes items describing eight situations that are designed to assess rejection anxiety and expectations. For each situation, respondents rate on a 6-point Likert scale their level of anxiety about the outcome (1 = *unconcerned*; 6 = *very concerned*) and the likelihood that the other person would respond in a cooperative manner (1 = *very unlikely*; 6 = *very likely*). A total score, ranging from 1 - 38, is derived by (a) multiplying anxiety scores by the reverse of the expectancy scores for each item and (b) summing the means of these products for each item. Higher scores indicate

more severe RS. When tested in a sample of undergraduate students, good internal consistency was observed ($\alpha = 0.82$; Gardner, Qualter, Stylianou, & Robinson, 2010). Comparable internal consistency was observed in present sample ($\alpha = 0.85$), with scores ranging from 1 - 33.38 ($M = 7.71$; $SD = 4.45$).

Procedure

Participation in this study took place online. After signing up for the study, participants were routed to a third-party website that hosted the survey. Before completing the battery of measures described above, informed consent was obtained from all participants. Following completion of the online survey, all participants were debriefed and granted research credit for a psychology course. All procedures were in accordance with the ethical standards of the institutional research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Results

Three violations of multivariate normality were detected during data screening. Despite acceptable skew, notable positive kurtosis was detected for PAS (kurtosis = 7.02), RSQ (kurtosis = 7.40), and PB (kurtosis = 10.28). Scores for each of these variables that were identified as univariate outliers were smoothed to represent no more than three standard deviations from the mean (Howell, 2013). Following the smoothing procedure, kurtosis values were consistent with a normal distribution for a sample this size (kurtosis < 5.16; Kline, 2005). The smoothed variables were utilized in all formal analyses. For means, standard deviations, and ranges of all variables used see **Table 1**.

To test primary hypotheses, we conducted two linear regression process models (Hayes, 2012) using the Hayes (2013) PROCESS macro in SPSS version 22. First, tests of indirect and conditional paths in PROCESS were obtained from coefficients estimating 1) the unique effect of TB on brooding, and 2) the unique effects of the initial predictors, brooding, RS, and the interaction between TB and RS on psychological pain (see **Figure 1**). Coefficients in the initial stage of this model identified a statistically reliable relationship between TB and brooding ($\beta = .577, p = < .001; sr^2 = .333$).

Coefficients estimated in the second stage of the analyses indicated a statistically significant indirect effect of brooding ($b = .195; 95\% \text{ CI} = [.088, .337]$) on the relation between TB and psychological pain. A statistically reliable direct effect for TB on psychological pain was no longer present ($\beta = .136, p = .069; sr^2 = .010$), suggesting full mediation. After accounting for the variance attributable to the indirect effect of brooding, we found a significant conditional effect of RS ($\beta = .237, p = < .001; sr^2 = .075$) on the residual relation between TB and psychological pain. Formal significance testing of simple slopes revealed that at higher levels of RS, there was a statistically significant positive impact of TB on psychological pain ($b = -.038, p = .001; sr^2 = .042; 95\% \text{ CI} = [.154, .548]$; see **Figure 2** for graph of simple slopes). However, at low levels of RS, the residual relation between TB and psychological pain was nonsignificant ($b = -.038, p = .708; sr^2 = < .001; 95\% \text{ CI} = [-.240, .164]$). A direct effect of RS on psychological pain was also detected. A summary of results can be found in **Table 2** (model 1).

Second, tests of indirect and conditional paths in PROCESS were obtained from coefficients estimating 1) the unique effect of PB on brooding, and 2) the unique effects of the initial predictors, brooding, RS, and the interaction between TB and RS on psychological pain (see **Figure 3**). Coefficients in the initial stage of this model identified a reliable relationship between

PB and brooding ($\beta = .387, p = < .001; sr^2 = .150$). Coefficients estimated in the second stage analysis indicated a statistically significant indirect effect of brooding ($b = .258$; 95% CI = [.127, .444]) on the relation between PB and psychological pain. However, a statistically reliable direct effect of PB on psychological pain remained ($\beta = .260, p = < .001; sr^2 = .028$), indicating only partial mediation. After accounting for the variance attributable to the indirect effect of brooding, we did not find a significant interaction of RS on the residual relation between TB and psychological pain ($\beta = .084, p = .731; sr^2 = .007$), suggesting that the relation between PB and psychological pain does not differ as a function of RS. A direct effect of RS on psychological pain was also detected. A summary of results are found in **Table 2** (model 2).

Discussion

Despite the growing body of research investigating thwarted interpersonal needs (see Ma, Batterham, Calear, & Han, 2016 for a review) and psychological pain (Klonsky & May, 2015) as predictors of suicidal desire, little attention has been paid to the connection between these constructs. In the current study, we found reliable relationships between both TB and PB and psychological pain. In addition, brooding acted as an intervening variable in these relationships and RS moderated the relation between TB and psychological pain.

Our finding of predictive associations between TB, and PB on psychological pain replicates the findings of Campos and Holden (2015) and extends Shneidman's (1993) theory by explaining two pathways, TB and PB, by which psychological pain may develop. In addition, this study provides an initial step in understanding how two major theories of suicide, IPTS (Van Orden et al., 2010) and Shneidman's theory of psychache (1993) may work symbiotically to produce suicidal inclinations.

As expected, RS directly predicted psychological pain. Although no prior research has examined the relationship between RS and psychological pain, these results are consistent with the finding that retrospective accounts of parental rejection predict psychological pain, as well as suicide risk in college students (Campos & Holden, 2015). Early experiences of rejection predict increased RS in youths (Wang, McDonald, Rubin, & Laursen, 2012; Rosenbach & Renneberg, 2014). Thus, our results may suggest RS as a possible pathway between thwarted interpersonal needs, such as rejection or isolation, and psychological pain.

Similarly, RS moderated the relation between TB and psychological pain, but not PB and psychological pain. This finding supports our conjecture that TB is affected by a proclivity to perceive rejection in social interactions. TB can be a consequence of objective (e.g., living alone or in a rural area) or subjective (e.g., feeling as though one does not belong in a family or social unit) social isolation. RS increases the risk of both outcomes, as those high in RS are more likely to recognize or misperceive signals relevant to rejection (Downey & Feldman, 1996). This heightened threat sensitivity increases the likelihood they believe they are being dismissed or rejected whether or not that is the case. As such, individuals high in RS often withdraw preemptively or become defensive, resulting in objective damage to social relationships (Downey & Feldman, 1996). Thus, it makes theoretical sense that RS exacerbates the relation between TB and psychological pain. Conversely, RS should not affect the relation between PB and psychological pain. PB describes the perception that one is a liability to his or her surrounding loved ones. We would not expect that being overly sensitive to rejection cues would exacerbate PB and lead to psychological pain.

Given its conceptual relevance and the paucity of research examining RS in the suicide literature, investigators should continue to examine the relation between RS and suicide related

constructs (e.g., TB, PB, psychological pain, suicide risk). One relevant avenue might be to examine how relational value, or the extent to which one believes social relationships are important, changes the relationship between TB, RS, and psychological pain. Wang et al. (2012) found that even at high levels of peer rejection, RS did not develop in middle school students low in relational value, suggesting that individuals who place high value in being accepted socially might be at greater risk for RS-related difficulties. Further, RS is viewed as a dispositional factor. Although there is a large literature examining dispositional factors as they relate to suicide ideation or attempts (May & Klonsky, 2016), these predictive tests are often not based in theory. As ideation-to-action frameworks (Klonsky & May, 2014) become the leading theoretical models of suicidal behavior, it will be important to understand how trait-based factors, like RS, relate to components of these models, such as interpersonal needs (Van Orden et al., 2010) or psychological pain (Klonsky & May, 2015).

As predicted, brooding acted as an intervening variable in the relations between PB and TB and psychological pain. Consistent with cognitive theories of psychopathology (e.g., Beck, Rush, Shaw, & Emery, 1979), these results suggest that it is not the mere existence of TB or PB that lead to psychological pain, but the way individuals think about and cope with these perceptions that leads to psychological pain. Brooding describes a passive and persistent state of repetitive negative thoughts and emotions and is often experienced as uncontrollable and inescapable (Kerkhof & van Spijker, 2011). The use of brooding to cope therefore intuitively increases psychological pain, as it involves rehearsing mental anguish, rather than using more adaptive emotion regulation strategies to instrumentally change the individual's situation (Treyner et al., 2003). Therefore, relative to individuals who have thoughts about being a burden or not belonging and cope with them through other means, individuals who engage in brooding develop

psychological pain. These findings are consistent with the research linking brooding to related adverse outcomes including depression (Treynor et al., 2003), nonsuicidal self-injury (Cohen et al., 2015), and suicide ideation and attempts (O'Connor, O'Connor, & Marshall, 2007).

Notably, brooding only partially mediated the relation between PB and psychological pain, suggesting that even without brooding, perceiving oneself as burdensome may still lead to psychological pain. PB has been described as a more reliable predictor of suicidal desire than TB (Ma et al., 2016). This assertion may provide context for the finding of a partial mediation and suggest that perceiving oneself as a burden is overtly detrimental even in the absence of dwelling or thinking repetitively about one's PB.

We chose to include brooding in this model based on its conceptual relevance to depression and consistent association with suicide-related outcomes in the literature (Rogers & Joiner, 2017). However, RST asserts that rumination is multifaceted and includes other types of cognitive responses, including reflection. Unlike brooding, reflection is neutrally valenced and involves contemplating one's situation to better understand and address one's depressive symptoms. Although the literature linking reflection to suicide ideation and attempts has been inconsistent (Morrison & O'Connor, 2008), it has been asserted that reflection is important to consider in the prediction of suicide, as it covaries less with depressive symptoms than brooding and may capture a unique facet of the rumination-ideation relationship. For example, one study found that reflection predicted suicidal ideation in individuals with previous suicide attempts above brooding (Surrence, Miranda, Marroquin, & Chan, 2009). Future research may benefit from integrating reflection into the current model. Particularly, it may be helpful to see if reflection better accounts for the relation between PB and psychache relative to brooding.

Clinically, these results offer two major approaches to reduce psychological pain in the presence of thwarted interpersonal needs. First, psychotherapy could work to decrease RS to reduce psychological pain. Interpersonal exercises, such as role-plays, can model actual versus perceived cues of rejection, as well as how to appropriately respond to these cues. Cognitive restructuring may also help clients arrive at accurate judgments about their interpersonal abilities and level of social support objectively available. Second, therapists may assess whether the client uses brooding as a primary form of coping with social difficulties. If so, therapists can use techniques such as cognitive restructuring to examine the validity and utility of brooding-related cognitions. Simultaneously, active forms of coping, such as behavioral activation and problem solving, can be introduced to reduce psychological pain by establishing concrete methods of bolstering social support.

There are several noteworthy limitations of this research. First, the sample used was largely homogenous, restricting the generalizability of these results to more diverse samples. Secondly, the cross-sectional design of this study precludes us from making assertions regarding the temporal nature of our results. For example, an alternate conception of our model might consist of psychological pain predicting thwarted interpersonal needs. Although we cannot rule out this possibility, we believe it is unlikely. Psychological pain is a broad construct that can describe any type of mental anguish, while IPTS constructs are restricted to the social domain. Consistent with Coyne's (1976) theory of depression in which interpersonal difficulties exacerbate depressive symptoms, we assert that it is more likely for thwarted interpersonal needs to act as one pathway to the larger construct of psychological pain, rather than global psychological pain leading to thwarted interpersonal needs. However, the most likely notion is that these constructs represent a reciprocal process in which psychological pain also makes individuals less likely to attempt to

connect with others, exacerbating TB or PB and perpetuating the cycle. Finally, this study used a nonclinical sample of college students and did not include a specific metric of suicide risk, limiting our ability to comment on the model's utility in predicting suicidal thoughts or behaviors. However, given the extensive literature tying IPTS constructs and psychological pain to suicide risk, we made this decision to more clearly focus on the conditional and indirect effects of RS and brooding on the theoretically based independent and dependent variables of interest, thereby establishing a conceptual base on which to build when including an estimate of suicide risk. We believe the next steps in this line of research are to test this model using suicide ideation or a cumulative measure of suicide risk as the outcome variable.

The current research provided an initial step in the simultaneous examination of dispositional and cognitive correlates in the relations between TB, PB, RS, and psychological pain. Although suicide risk is multideterminant, this model represents one method of integrating and extending leading theories of suicide to include meaningful therapeutic targets. Future studies should extend this research to test the assumptions of this model in samples at elevated risk for suicidal ideation and attempts.

References

- Beck, A. T., Rush, A. J., Shaw, B. F., & Emery, G. (1979). *Cognitive therapy of depression*. New York: The Guilford Press.
- Breines, J. G., & Ayduk, O. (2015). Rejection sensitivity and vulnerability to self-directed hostile cognitions following rejection. *Journal of Personality*, 83, 1–13.
- Bryan, C. J., Morrow, C. E., Anestis, M. D., & Joiner, T. E. (2010). A preliminary test of the interpersonal-psychological theory of suicide in a military sample. *Personality and Individual Differences*, 48, 347–350.
- Campos, R. C., & Holden, R. R. (2015). Testing models relating rejection, depression, interpersonal needs, and psychache to suicide risk in nonclinical individuals. *Journal of Clinical Psychology*, 71, 994–1003.

Chesin, M. S., & Jeglic, E. L. (2016). Factors associated with recurrent suicidal ideation among racially and ethnically diverse college students with a history of suicide attempt: The role of mindfulness. *Archives of Suicide Research*, 20, 29–44.

Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Hillsdale, NJ: Erlbaum.

Cohen, N. J., Stange, J. P., Hamilton, J. L., Burke, T. A., Jenkins, A., ... Alloy, L. B. (2015). The interaction of affective states and cognitive vulnerabilities in the prediction of non-suicidal self-injury. *Cognition and Emotion*, 29, 539–547.

Coyne, J. C. (1976). Toward an interactional description of depression. *Psychiatry*, 39, 28–40.

Downey, G., & Feldman, S. I. (1996). Implications of rejection sensitivity for intimate relationships. *Journal of Personality and Social Psychology*, 70, 1327–1343.

Efron, B., & Tibshirani, R. J. (1994). *An introduction to the bootstrap*. Boca Raton, FL: CRC Press.

Gardner, K. J., Qualter, P., Stylianou, M., & Robinson, A. J. (2010). Facial affect recognition in non-clinical adults with borderline personality features: The role of effortful control and rejection sensitivity. *Personality and Individual Differences*, 49, 799–804.

Hayes, A. F. (2012). *PROCESS: A versatile computational tool for observed variable mediation, moderation, and conditional process modeling [White paper]*. Retrieved from <http://www.afhayes.com/public/process2012.pdf>

Hayes, A. F. (2013) *Introduction to mediation, moderation, and conditional process analysis: A Regression-based approach*. New York: Guilford Publications.

Holden, R. R., Mehta, K., Cunningham, E. J., & McLeod, L. D. (2001). Development and preliminary validation of a scale of psychache. *Canadian Journal of Behavioural Science*, 33, 224–232.

Joiner, T. E. (2005). *Why people die by suicide*. Cambridge, MA: Harvard University Press.

Joiner, T. E., Riberio, J. D., & Silva, C. (2012). Nonsuicidal self-injury, suicidal behavior, and their co-occurrence as viewed through the lens of the Interpersonal Theory of Suicide. *Current Directions in Psychological Science*, 21, 342–347.

Kerkhof, A. & van Spijker, B. (2011). Worrying and rumination as proximal risk factors for suicidal behaviour. In R. C. O'Connor, S. Platt, & J. Gordon (Eds.), *International handbook of suicide prevention research, policy, and practice* (pp. 199–210). Malden, MA: John Wiley & Sons.

Kline, R. B. (2005). *Principles and practice of structural equation modeling* (2nd ed.). New York: Guilford Press.

Klonsky, E. D., & May, A. M. (2014). Differentiating suicide attempters from suicide ideators: A critical frontier for suicidology research. *Suicide and Life-Threatening Behavior*, 44, 1–5.

- Klonsky, E. D., & May, A. M. (2015). The three-step theory (3ST): A new theory of suicide rooted in the “ideation-to-action” framework. *International Journal of Cognitive Therapy*, 8, 114–129.
- Levy, S. R., Ayduk, O., & Downey, G. (2001). The role of rejection sensitivity in people’s relationships with significant others and valued social groups. In M. R. Leary (Ed.), *Interpersonal rejection* (pp. 293–315). New York: Oxford University Press.
- London, B., Downey, G., Bonica, C., & Paltin, I. (2007). Social causes and consequences of rejection sensitivity. *Journal of Research on Adolescence*, 17, 481–506.
- Ma, J., Batterham, P. J., Caele, A. L., Han, J. (2016). A systematic review of the predictions of the Interpersonal-Psychological Theory of suicidal behavior. *Clinical Psychology Review*, 46, 34–45.
- MacKinnon, D. P., Lockwood, C. M., Hoffman, J. M., West, S. G., & Sheets, V. (2000). A comparison of methods to test mediation and other intervening variable effects. *Psychological Methods*, 7, 83–104.
- May, A. M., & Klonsky, E. D. (2016). What distinguishes suicide attempters from suicide ideators? A meta-analysis of potential factors. *Clinical Psychology: Science and Practice*, 23, 5–20.
- Morrison, R., & O’Connor, R. C. (2008). A systematic review of the relationship between rumination and suicidality. *Suicide and Life-Threatening Behavior*, 38, 523–538.
- Nock, M. K., Prinstein, M. J., & Sterba, S. (2009). Revealing the form and function of self-injurious thoughts and behaviors: A real-time ecological assessment study among adolescents and young adults. *Journal of Abnormal Psychology*, 118, 816–827.
- Nolen-Hoeksema, S. (1991). Responses to depression and their effects on the duration of depressive episodes. *Journal of Abnormal Psychology*, 100, 569–582.
- O’Connor, D. B., O’Connor, R. C., & Marshall, R. (2007). Perfectionism and psychological distress: Evidence of the mediating effects of rumination. *European Journal of Personality*, 21, 429–452.
- O’Connor, R. C., & Noyce, R. (2008). Personality and cognitive processes: Self-criticism and different types of rumination as predictors of suicidal ideation. *Behavior Research & Therapy*, 48, 392–401.
- Rogers, M. L., & Joiner, T. E. (2017). Rumination, suicidal ideation, and suicide attempts: A meta-analytic review. *Review of General Psychology*, 21, 132–142.
- Rosenbach, C., & Renneberg, B. (2014). Rejection sensitivity as a mediator of the relationship between experienced rejection and borderline characteristics. *Personality and Individual Differences*, 69, 176–181. doi:10.1016/j.paid.2014.05.032
- Rueger, S. Y., Malecki, C. K., Pyun, Y., Aycock, C., & Coyle, S. (2016). A meta-analytic review of the association between perceived social support and depression in childhood and adolescence. *Psychological Bulletin*, 142, 1017–1067.

- Shneidman, E. S. (1993). Suicide as psychache, *Journal of Nervous and Mental Disease*, 181, 145–157.
- Surrence, K., Miranda, R., Marroquin, B. M., & Chan, S. (2009). Brooding and reflective rumination among suicide attempters: Cognitive vulnerability to suicidal ideation. *Behavior Research and Therapy*, 47, 803–808.
- Tossani, E. (2013). The concept of mental pain. *Psychotherapy and Psychosomatics*, 82, 67–73.
- Treynor, W., Gonzalez, R., Nolen-Hoeksema, S. (2003). Rumination reconsidered: A psychometric analysis. *Cognitive Therapy and Research*, 27, 247–259.
- Troister, T., & Holden, R. R. (2010). Comparing psychache, depression, and hopelessness in their associations with suicidality: A test of Shneidman's theory of suicide. *Personality and Individual Differences*, 49, 689–693.
- Van Orden, K. A., Witte, T. K., Cukrowicz, K. C., Braithwaite, S. R., Selby, E. A., & Joiner, T. J. (2010). The interpersonal theory of suicide. *Psychological Review*, 117, 575–600.
- Wang, J., McDonald, K. L., Rubin, K. H., & Laursen, B. (2012). Peer rejection as a social antecedent to rejection sensitivity in youth: The role of relational valuation. *Personality and Individual Differences*, 53, 939–942.
- Watson, J., & Nesdale, D. (2012). Rejection sensitivity, social withdrawal, and loneliness in young adults. *Journal of Applied Social Psychology*, 42, 1984–2005.

Table 1. Means, Standard Deviations, and Ranges for Measures Used in Study.

Variable	M (SD)	Range
PAS	17.80 (8.94)	13 – 64
TB	12.85 (7.21)	5 – 35
PB	7.01 (4.49)	5 – 34
RRS-B	9.33 (3.70)	5 – 20
RSQ-SF	7.71 (4.45)	1 – 33

Note: *M* = Mean; *SD* = Standard Deviation; PAS = Psychache Scale; TB = thwarted belongingness subscale of Interpersonal Needs Questionnaire; PB = perceived burdensomeness subscale of Interpersonal Needs Questionnaire; RRS-B = Brooding subscale of Ruminative Response Scale; RSQ-SF = Rejection Sensitivity Questionnaire – Short Form.

Table 2. Coefficients, CIs, and effect sizes for indirect effect of brooding and conditional effect of RS on the relations between TB, PB, and psychache.

DV	IV	β	b (SE)	95% CI	p -value	pr	sr^2
Model 1: TB							
RRS-B (α path)							
$R^2 = .333$	TB	.577	.296 (.040)	[.229, .363]	< .001	.577	.333
PAS (β path)							
$R^2 = .500$	TB	.136	.154 (.084)	[-.012, .320]	.069	.149	.010
RRS-B	.299	.657 (.159)	[.343, .972]	< .001	.322	.052	
RSQ	.222	.419 (.140)	[.143, .695]	.003	.246	.029	
RSQ*TB	.237	.046 (.013)	[.020, .071]	< .001	.378	.075	
$\alpha\beta$ path							
RRS-B		.195 (.063)	[.088, .337]				
Model 2: PB							
RRS-B (α path)							
$R^2 = .150$	PB	.387	.347 (.067)	[.215, .478]	< .001	.387	.150

PAS (β path)							
$R^2 = .548$	PB	.260	.681 (.160)	[.365, .996]	< .001	.244	.028
RRS-B	.364	.743 (.138)	[.471, 1.02]	< .001	.416	.093	
RSQ	.217	.355 (.127)	[.104, .606]	.006	.253	.030	
RSQ*PB	.084	.009 (.025)	[-.040, .057]	.731	.125	.007	
$\alpha\beta$ path							
RRS-B		.258 (.079)	[.127, .444]				

Note: CI = confidence interval; DV = dependent variable; IV = independent variable; RS = rejection sensitivity; SE = Standard Error; 95% CI = 95% Confidence interval for b ; pr = partial correlation; sr^2 = squared semi-partial correlation; TB = thwarted belongingness subscale of Interpersonal Needs Questionnaire; PB = perceived burdensomeness subscale of Interpersonal Needs Questionnaire; RRS-B = Brooding subscale of Ruminative Response Scale; RSQ = Rejection Sensitivity Questionnaire; Bolded coefficients indicate a statistically significant effect.

Figure 1. Indirect and conditional effects of brooding and rejection sensitivity on the relation between TB and psychological pain. *Note:* Path coefficients reflect standardized effects (β); * indicates $p < .01$. Model equations: Brooding = $a + b_1(TB) + e_i$; Psychological Pain = $a + b_3(TB) + b_4(Brooding) + b_5(Rejection) + b_6(TB*Rejection) + e_i$.

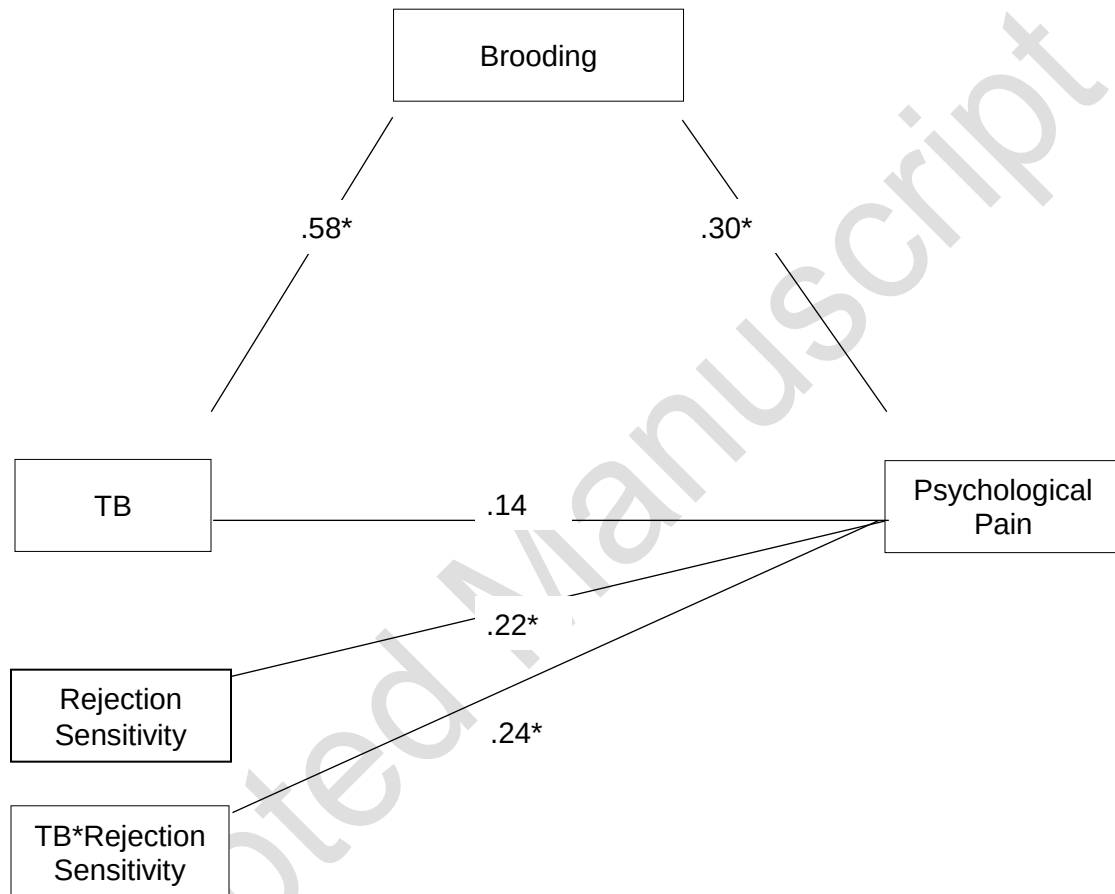


Figure 2. Unstandardized simple slopes for the interaction of rejection sensitivity and TB on psychological pain. *Note:* TB = thwarted belongingness subscale of Interpersonal Needs Questionnaire; RS = Rejection Sensitivity Questionnaire-Short Form; High TB consistent with + 1SD; Low TB consistent with -1 SD.

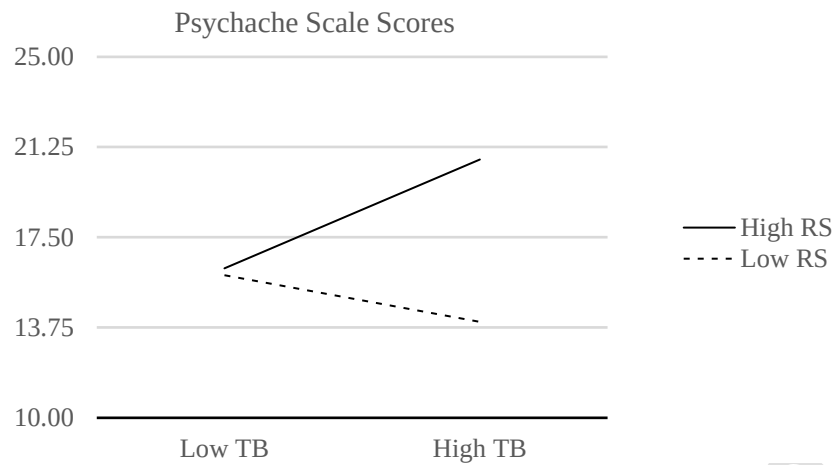


Figure 3. Indirect and conditional effects of brooding and rejection sensitivity on the relation between PB and psychological pain. *Note:* Path coefficients reflect standardized effects (β); * indicates $p < .01$. Model equations: Brooding = $a + b_1(\text{PB}) + e_i$; Psychological Pain = $a + b_3(\text{PB}) + b_4(\text{Brooding}) + b_5(\text{Rejection}) + b_6(\text{PB} \times \text{Rejection}) + e_i$.

